

leaderboard and specializes in domain-specific embeddings across science, legal, finance, and coding. While OpenRouter doesn't publicly list its pricing, this model uniquely excels at cross-domain applications and multilingual tasks, making it valuable when working with specialized content or international applications.

Qwen's embedding family (qwen3-embedding-8b/4b/0.6b) stands out with 32,000 token context windows—4x longer than OpenAI models. Priced at \$0.02 per million tokens regardless of model size, these models offer exceptional value for long-document processing. The Qwen3 series supports 100+ languages with particularly strong Chinese/English performance, includes specialized training for code retrieval, and implements Matryoshka Representation Learning for flexible output dimensions. The 0.6B variant provides efficient processing for resource-constrained environments, while the 8B model delivers near-frontier performance at budget pricing.

Complete RAG implementation with authentication and setup

Setting up OpenRouter requires minimal configuration. Create an account at openrouter.ai, generate an API key from openrouter.ai/keys, and optionally set credit limits for cost management. The authentication uses standard bearer tokens compatible with OpenAI's SDK:

```
python

import os
from openai import OpenAI

# Initialize OpenRouter client
client = OpenAI(
    base_url="https://openrouter.ai/api/v1",
    api_key=os.environ.get("OPENROUTER_API_KEY"),
)
```